

KENLEY AF, United Kingdom

WMO: 037810

Lat: 51.3039N

Lon: 0.0914W

Elev: 558

StdP: 14.40

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.3	28.4	21.1	16.1	32.4	23.6	18.2	32.9	28.6	48.5	26.1	48.2	6.1	30	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	79.8	65.0	76.0	63.2	72.8	61.8	66.6	76.2	64.8	73.0	63.3	70.3	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	88.6	69.3	61.8	84.2	67.8	60.4	80.1	66.5	31.4	76.3	30.1	73.1	29.0	70.3	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
23.0	20.4	18.1	DB	22.4	85.4	3.4	4.2	19.9	88.4	17.9	90.9	16.0	93.3	13.5	96.3	78.2
			WB	22.1	70.4	3.5	3.0	19.6	72.6	17.6	74.3	15.7	76.0	13.2	78.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	40.1	40.6	43.7	48.2	53.4	58.8	62.9	62.5	57.9	52.1	45.6	41.4
	DBStd	9.73	5.73	5.79	5.50	5.65	5.50	4.81	4.64	4.59	4.48	5.00	5.59	6.12
	HDD50	1356	309	265	202	96	25	1	0	0	1	35	151	271
	HDD65	5314	773	683	659	503	360	194	98	108	219	402	583	732
	CDD50	1596	1	2	8	44	131	266	401	386	237	98	18	4
	CDD65	80	0	0	0	0	1	10	34	29	5	1	0	0
	CDH74	535	0	0	0	2	11	76	233	188	23	2	0	0
	CDH80	87	0	0	0	0	0	11	37	38	1	0	0	0
Wind	WSAvg	8.7	10.3	10.1	9.2	8.4	8.3	8.0	7.9	7.6	7.5	8.5	8.8	9.7
Precipitation	PrecAvg	27.9	2.8	2.0	1.7	1.9	1.9	2.0	1.8	2.3	2.2	3.0	3.1	2.8
	PrecMax	39.1	7.3	4.6	4.1	5.1	4.2	5.4	4.1	4.3	4.8	6.8	6.5	6.0
	PrecMin	19.7	0.5	0.2	0.4	0.1	0.3	0.4	0.4	0.1	0.5	0.8	1.3	0.4
	PrecStd	5.0	1.5	1.2	0.9	1.3	1.0	1.4	0.9	1.1	1.2	1.5	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	56.5	61.9	71.5	75.5	81.2	84.2	84.2	77.1	68.0	59.2	55.3
		MCWB	51.0	49.6	51.6	57.0	60.8	65.5	66.6	66.7	63.4	60.5	56.7	53.0
	2%	DB	51.9	52.3	57.3	65.1	71.1	75.5	80.0	79.4	72.0	63.2	57.0	53.7
		MCWB	49.9	48.8	49.8	54.0	59.3	62.6	65.0	65.2	62.2	58.1	54.7	51.9
	5%	DB	50.3	50.4	54.3	60.8	67.3	71.7	76.3	74.6	68.2	61.0	55.2	52.2
		MCWB	48.4	47.4	48.5	51.5	57.5	61.0	63.5	63.1	60.3	57.2	52.9	50.4
	10%	DB	48.5	48.5	51.8	57.4	63.3	68.3	72.7	70.8	65.4	59.3	53.6	50.6
		MCWB	46.7	46.0	47.5	49.7	55.0	59.2	61.9	61.5	58.7	56.2	51.3	48.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	51.6	53.1	58.5	63.9	67.5	69.3	69.0	66.2	61.9	57.5	53.8
		MCDB	52.9	54.0	58.1	69.3	71.7	78.5	81.0	80.7	73.2	66.0	58.4	54.6
	2%	WB	50.5	49.8	51.2	55.1	60.9	64.4	66.5	66.5	63.6	59.7	55.3	52.4
		MCDB	51.6	51.8	55.0	63.3	68.3	72.5	76.5	75.9	69.6	62.0	56.7	53.4
	5%	WB	48.6	48.1	49.9	52.7	58.4	62.2	64.7	64.6	61.9	58.2	53.5	50.8
		MCDB	49.9	49.8	53.2	58.7	65.4	69.5	73.7	71.8	66.4	60.3	55.0	52.1
	10%	WB	47.0	46.4	48.2	50.9	56.1	60.3	63.0	62.9	60.2	56.8	51.7	48.8
		MCDB	48.4	48.3	51.4	56.0	62.2	66.9	70.5	69.1	63.9	58.8	53.3	50.4
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.0	11.2	13.8	14.2	14.6	14.9	14.2	13.2	10.6	8.9	8.0
		MCDBR	9.4	10.3	14.9	18.4	19.8	20.1	20.8	19.4	16.7	11.6	8.6	8.1
		MCWBR	9.6	8.0	9.3	10.1	10.8	10.4	9.4	8.8	8.8	7.9	7.8	8.3
	5% WB	MCDBR	9.4	9.6	12.8	16.5	18.0	18.0	18.1	16.8	14.4	10.2	8.5	8.2
		MCWBR	9.8	8.6	9.0	9.7	10.4	10.4	9.5	8.7	9.0	8.2	8.2	8.6
Clear-Sky Solar Irradiance	taub	0.324	0.340	0.375	0.394	0.399	0.399	0.395	0.387	0.370	0.354	0.328	0.314	
	taud	2.388	2.395	2.289	2.256	2.283	2.305	2.321	2.364	2.409	2.438	2.432	2.373	
	Ebn at Noon	218	248	259	267	272	272	271	267	259	240	216	201	
	Edh at Noon	20	25	34	39	40	39	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	257	470	827	1304	1566	1736	1648	1373	1029	591	307	200	
	RadStd	22	64	108	134	103	158	149	107	82	53	35	25	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page